

# What a low-cost smartphone protocol can and cannot establish about urban noise

Hanoi, 363 field measurements, three negative results  
and the instrument we ended up building

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June – August 2026

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1. Context and research questions
2. Data and protocol
3. Method and evaluation protocol
4. Results
5. Auditing our own data
6. From a pipeline to an instrument
7. Next: Clermont-Ferrand

Everything on these slides is reproducible from [github.com/Colauz/noise-modelling-hanoi](https://github.com/Colauz/noise-modelling-hanoi)

## Context and research questions

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## The gap this project sits in

- Hanoi: dense, fast-growing, **motorcycle-dominated** traffic
- **No national strategic noise-mapping programme** and no statutory noise action plan in Vietnam

Nguyen et al. 2025, *City Environ. Interactions*

- **One** instrumented campaign ever published for Hanoi — and it measured in **2005–2007**

Phan et al. 2010, *Applied Acoustics* 71(5)

- Class 1 sound level meters and commercial propagation software: out of reach for most local research budgets

### The question

How far does a protocol built from **three consumer smartphones** and **open data** actually get you?

The honest answer turned out to be: further than nothing, and much less far than a map.

## The research question, and what is out of scope

### The question, as we posed it in June — unchanged

Can urban noise be attributed to **the form of the city**, and to **the type of vehicle** passing through it? Models validated on car-centric European cities are unproven on motorcycle-heavy soundscapes.

### How it was operationalised

1. Can OpenStreetMap morphology predict noise at an **unmeasured location**?
2. Does a model **transfer** between cities?
3. Does vehicle **flow from video** explain the levels?
4. What can the protocol claim with **no absolute reference**?

**Out of scope, declared up front** — night-time (10 points after 21:00), compliance against any standard, any prediction beyond the three sites.

### And the answer, on both halves

**Form:** **no** — morphology adds nothing the strict protocols can see.

**Vehicle type:** **no** — and the chain that was meant to measure it does not work.

What survives is  $\log(d_{\text{road}})$ : distance to the road, and little else.

## Related work — ten references, selected by what they are used for

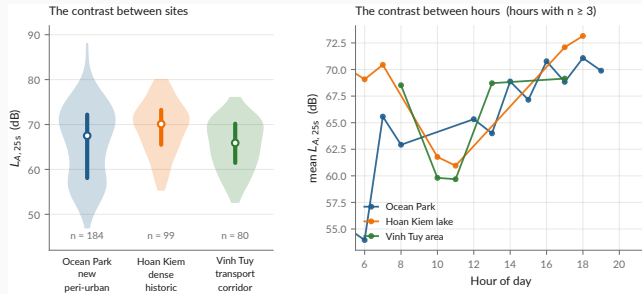
| Starting point                                                                                                                          | Method                                                                                    | Anchoring                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Sunbird Urban Noise Uganda<br>61K — 61k clips, <i>Sci. Data</i> 2026                                                                    | Roberts et al. 2017, <i>Ecography</i><br>— spatial CV must exclude the<br>feature support | Phan et al. 2010 — the only<br>instrumented Hanoi campaign |
| The dataset we set out to transfer from                                                                                                 | The paper that told us our first<br>$R^2$ was wrong                                       | The reference our levels are compared against              |
| <b>Also used:</b> CNOSSOS-EU · NoiseModelling · GAMA · Gelb & Apparicio 2019 (dosimeters, HCMC) · QCVN 26:2010/BTNMT · TCVN 7878-2:2010 |                                                                                           |                                                            |

Full annotated bibliography with verified DOIs: [docs/literature-review.md](#)

## Data and protocol

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# The campaign



data/processed/measurements.csv

**Protocol** — 3 handsets, *Decibel X* (A-weighted, SLOW), 20–30 s per point, a clip  $\geq 10$  s, a 10 m GPS gate, ODK Collect → Kobo.

**Published, CC-BY-4.0** — both CSVs and both XLSForms. Videos and raw Kobo exports are **not**: faces, plates, collector identities.



Three phones, cross-calibrated against *each other* and against nothing

A bias common to all three is **invisible in the data by construction**.

## What that forbids

- Stating an absolute level as a fact
- Any compliance claim under QCVN 26:2010 — it requires a class 1 or 2 instrument

## What it still allows

- Contrasts *between places* and *between hours* — a common bias cancels in a difference
- A *bounded* bias interval from instrumented literature: **+3.0 to +7.3 dB**

The measured quantity, and the only one:

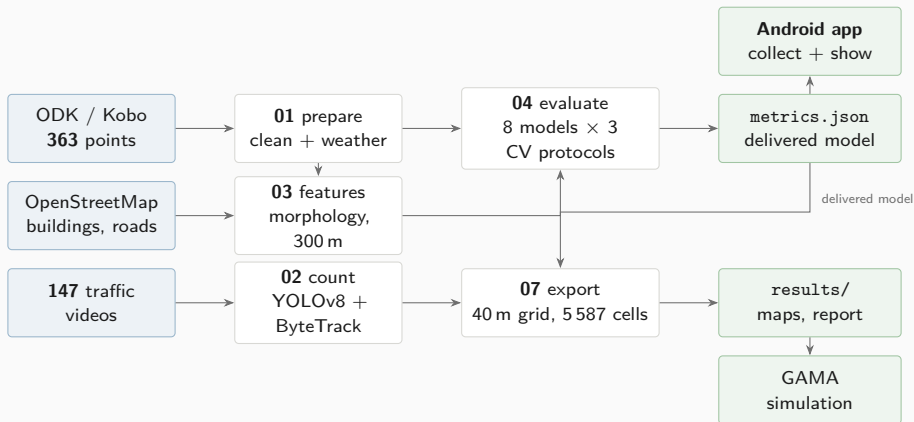
$$L_{A,25s} = 10 \log_{10} \left( \frac{1}{T} \int_0^T \frac{p_A^2(t)}{p_0^2} dt \right), \quad T \approx 25 \text{ s}, \quad p_0 = 20 \mu\text{Pa}$$

**Not**  $L_{Aeq,8h}$ , **not**  $L_{den}$ , **not**  $L_{night}$  — those integrate over hours or a year.

## Method and evaluation protocol

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# The chain, end to end



The delivered model is chosen **by code** — nothing downstream can silently substitute another.

`scripts/01_...11_ · docs/methodology.md`

# The evaluation protocol — and why it is the whole story

Every model is scored under **three splits**, on exactly the same folds:

1. **Block-CV, 600 m** — square blocks in UTM 48N, GroupKFold on the block. The permissive one.
2. **Buffered leave-one-out, 300 m** — for each point, train on *all* points more than 300 m away.

**The reference protocol.**

3. **Leave-one-site-out** — generalisation to an unseen urban typology.

Every score carries a 95 % interval bootstrapped **by block**, not by point: resampling correlated points gives an interval that is far too narrow.

## Why 300 m and not less

The morphology features aggregate over a disc of **radius 300 m**. Two points 110 m apart share more than **85 %** of that disc.

Group on 110 m cells and the model sees near-twins of its test points in training. The score is then not out-of-sample.

Roberts et al. 2017, *Ecography*

`scripts/04_evaluate_models.py`

## Eight models, three baselines, one ablation

Nothing is compared against nothing. The baselines are the point:

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**Baselines** — what any model must beat to have said anything

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|                |                                                                |
|----------------|----------------------------------------------------------------|
| global_mean    | the absolute floor                                             |
| site_mean      | no fine spatial information at all                             |
| site_hour_mean | mean per (site, hour) — <b>the one to beat</b>                 |
| dist_road      | linear regression on $\log(d_{\text{road}})$ — minimal physics |
| idw            | inverse distance weighting — pure interpolation                |

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**Ablation** — which half of the feature set is doing the work

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|             |                     |
|-------------|---------------------|
| lgbm_time   | hour + weekend only |
| lgbm_morpho | morphology only     |
| lgbm_full   | both — the v1 model |

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**V2** — feature engineering, physics, and the architecture we had recommended

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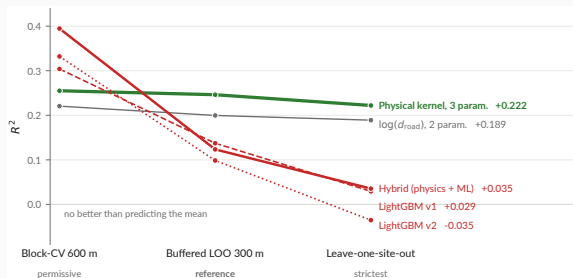
|          |                                                                     |
|----------|---------------------------------------------------------------------|
| lgbm_v2  | distances split by road class, cyclic hour                          |
| physical | line-source kernel, three positive parameters, <i>no morphology</i> |
| hybrid   | physical kernel + LightGBM on the residual                          |

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# Results

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# The ranking inverts — and that is the result



`models/metrics.json -- the same folds for every model`

## Read across, not down

The elaborate architecture **wins the permissive split and loses both strict ones.**  
Three parameters is the only flat line.

## What we did about it

The hybrid is **tested and rejected at this sample size** — not deferred to future work.

## The delivered model

A **line**-source attenuation kernel — energy falling as  $1/d$ , not  $1/d^2$ : a street is a line of sources, not a point.

$$E = \frac{A_{\text{hw}}}{\max(d_{\text{hw}}, D_0)} + \frac{A_{\text{res}}}{\max(d_{\text{res}}, D_0)} + B$$
$$L = 10 \log_{10} E$$

$$A_{\text{hw}} = 4.774 \times 10^7 \quad A_{\text{res}} = 3.800 \times 10^7 \quad D_0 = 5 \text{ m}$$

- $-3 \text{ dB}$  per doubling of distance, not  $-6$
- Positivity-constrained:  $A_{\text{hw}}, A_{\text{res}}, B > 0$
- **$B$  fits to  $-99.6 \text{ dB}$** , a boundary solution. **The background term is not identifiable here.**

`models/hybrid_physical.json` · `models/metrics.json`

### Performance under the reference protocol

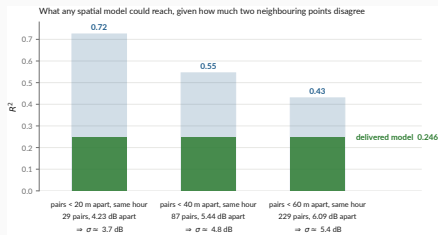
|                          |                |
|--------------------------|----------------|
| $R^2$                    | <b>0.246</b>   |
| 95 % CI                  | [0.097, 0.339] |
| RMSE                     | 6.20 dB        |
| MAE                      | 5.01 dB        |
| <hr/>                    |                |
| data spread ( $\sigma$ ) | 7.14 dB        |
| <hr/>                    |                |

### Say the error out loud

A typical prediction is wrong by **5 dB** — a factor of three in energy. Enough for a contrast, nowhere near an assessment.



“ $R^2 = 0.25$  — isn't that very low?”



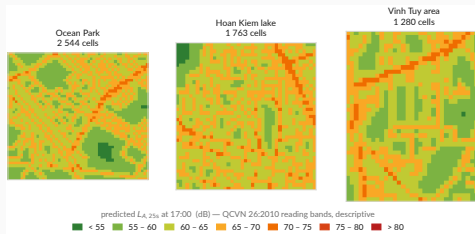
data/processed/measurements.csv · docs/metrology.md

Two points a few metres apart *at the same hour* must be predicted identically. Whatever they disagree by is variance **no spatial model can explain**:  $\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2} \mathbb{E}|\Delta L|$ , and  $R_{\max}^2 = 1 - \sigma^2/s^2$ .

At the 40 m scale the map works at

The ceiling is near  $R^2 \approx 0.54$ . The delivered **0.246** is about **half** of what is attainable, not a tenth.

## The map — and the caveat that goes with it



results/maps/hanoi\_noise\_map.csv -- 5587 cells

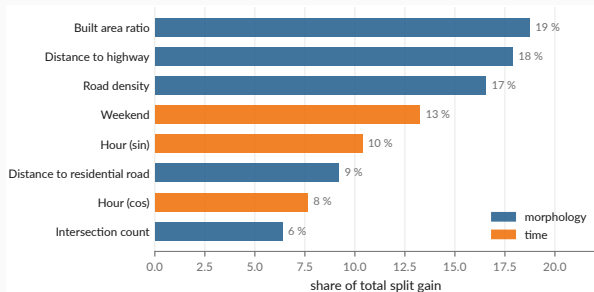
### The grid step is not the accuracy

40 m cells from a model whose typical error is **5 dB**. Bands are QCVN 26:2010, **descriptive**.

### And no hour, either

17 hourly columns, **all 17 identical**: the kernel has no hour term. **Nothing varies that the model does not model.**

## What the learned model was leaning on



`models/metrics.json : feature_importance_gain`

Two thirds of LightGBM's split gain goes to **morphology** — built area, road density, distance to the two road classes.

### And it buys nothing

Morphology alone scores  $R^2 = 0.041$  under the reference protocol. **Where a model spends its capacity and where it earns its score are different questions**, and only the second one is a result.

This is why the ablation is on the same three splits as everything else.

# The three negative results

## 1 — A three-parameter physical law beats every learned model we built

Six-variable gradient boosting, engineered features, a hybrid architecture: all lose the reference protocol to two distances and a constant. At  $n = 363$  over three sites, there is not enough signal to justify capacity.

## 2 — Cross-city transfer fails, and in three distinct ways

Pretrained on Sunbird Uganda, scored on our 363 points:  $R^2 = -15.9$  (v1),  $-8.2$  (convention-invariant v2). **Not a calibration offset** — removing the 26 dB mean difference still leaves  $R^2$  negative. **v1 transfers anti-information**,  $r = -0.38$ : it ranks quiet and loud the wrong way round. **v2 leaves nothing**,  $r = +0.08$  — not wrong, empty. Against which  $\log(d_{\text{road}})$  alone reaches **0.240** on the same points.

## 3 — Vehicle flow does not explain the levels

Non-negative energy regression on 147 matched videos returns **zero coefficients** for motorcycles and cars. Speed and source–receiver distance are not observable from a non-georeferenced camera. **Section 4 revisits how much of this is physics and how much is instrument.**

```
make transfer · scripts/experiments/uganda_transfer.py · docs/negative-results.md
```

## Auditing our own data

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## We found one of our own published results was wrong

### Retracted, August 2026

A LightGBM model scored  $R^2 = 0.45$  under what was then described as honest spatial cross-validation. A GAMA simulation was built on it. A project-update deck presented the figure.

The grouping was on **110 m cells** while the features aggregate over **300 m**. It leaked.

### What replaced it

$R^2 = 0.246$  under buffered leave-one-out — roughly half the withdrawn figure, and the one that survives a split excluding the feature support.

- The retracted deck and map are in docs/archive/, **with their reason**, not deleted
- Nine corrections applied the same day
- The project pivoted from *producing a noise map* to *studying what this protocol supports*

`docs/audit/scientific-audit.md` · `docs/project-timeline.md` · `docs/archive/`

## A second audit, on the data itself

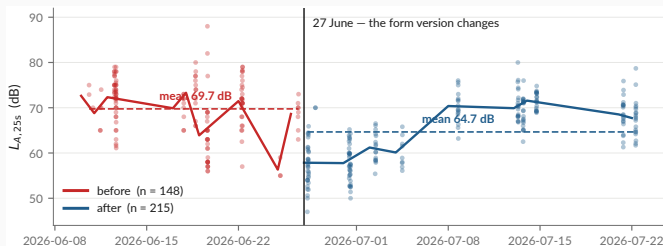
The first audit checked the *method*. This one checks the *measurements*.

One question, over the published CSVs, the raw Kobo export and the cleaning script: **does anything here behave in a way the documentation does not explain?**

| Checked                                          | Result                        |         |
|--------------------------------------------------|-------------------------------|---------|
| Raw rows vs published rows                       | 363 → 363, none dropped       | clean   |
| Every README figure vs <code>metrics.json</code> | exact match, none hand-copied | clean   |
| Coordinate duplicates                            | none at 1 m                   | clean   |
| Weather fields                                   | consistent hourly API values  | clean   |
| Test suite                                       | 34 passing                    | clean   |
| Level distribution over time                     | a discontinuity on 27 June    | finding |
| Video→measurement chain                          | modal split implausible       | finding |
| Confidence intervals in the headline table       | overlapping                   | finding |

The three findings are on the next two slides. None was known when the results section was written.

## Finding 1 — a $-5.1$ dB discontinuity on 27 June



data/processed/measurements.csv · docs/audit/

Controlling for site, hour and class:

$-5.14$  dB (SE 0.55,  $p \approx 10^{-20}$ ). The form version changes the same day — integer readings 87 %  $\rightarrow$  20 %.

### Why it matters

Before/after is predictable from (lat, lon) alone at **AUC 0.83**. An instrumental shift is then spatially structured, and a model can learn it **as morphology**. Status: **open**.



## Findings 2 and 3 — the video chain, and the intervals

### 2 — The video chain is measuring something, but not what we think

- Detector modal split: **57 % cars, 36 % motorcycles**. In Hanoi. Reality is roughly the inverse.
  - Matched subset spans 61–80 dB against 47–88 dB overall:  $\sigma$  cut by 42 %
  - Flow vs level:  $r = -0.11$ . A *negative* correlation between traffic and noise is not physical.
- Video→measurement gaps: median 15 s, p90 56 s, max 161 s — for a 25 s sample

**Consequence for negative result 3:** it is far more likely an **instrument failure** than a finding about acoustics. It should be reported as *“the chain could not be made to work”*.

### 3 — The headline ranking is not statistically separated

physical  $R^2 = 0.246$ , CI [0.097, 0.339] vs dist\_road  $R^2 = 0.200$ , CI [0.077, 0.266]. The intervals overlap almost entirely. **“The physical kernel beats log-distance”** is a coin flip, and the deck’s own table should show the CIs.

## Holds

- The evaluation protocol and the ranking inversion
- Negative result 1 — capacity does not pay at  $n = 363$
- Negative result 2 — transfer fails,  $R^2 < 0$
- Provenance: raw  $\rightarrow$  published is lossless and checkable
- Every published number traces to one file

## Needs work before submission

- The 27 June discontinuity — diagnose, then refit with the phase
- Negative result 3 — reframe as an instrument failure
- Carry the confidence intervals into every summary table
- Flag observed vs derived values in the published CSV (15 % of `construction_nearby` is inferred from the noise class)

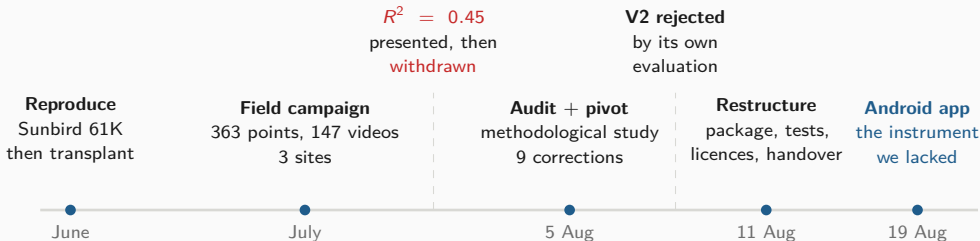
**The instrument was the problem in July, and it is still the problem in August.**

Which is the argument for the next section.

## **From a pipeline to an instrument**

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# Three months, in one picture



## The pattern

Every phase ended by **rejecting its own preferred answer**: transfer, then the leaked  $R^2$ , then the hybrid architecture we had recommended ourselves.

## Where that led

Each rejection pointed at the same root cause — an instrument we did not control. So the last three weeks were spent building one.

`docs/project-timeline.md`

# Why build an application at all

## Four things the campaign could not do

1. **Control the instrument.** *Decibel X* is closed: no visible weighting, window or gain.
2. **Measure and record in one session.** Two apps meant the level of one stretch of street and the audio of another.
3. **Scale beyond three people.** Nobody pairs ODK Collect and a sound meter by hand.
4. **Record what the device was.** Handset, OS, audio source — none of it reached the dataset.

### The decision, 19 August 2026

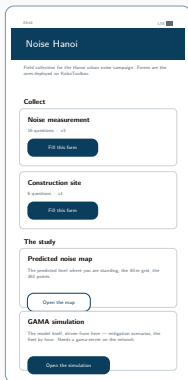
One application that **replaces the pairing** — forms, level, clip, GPS fix, submission, offline-first — and **carries the study**, negative results included.

mobile/README.md   Kotlin + Compose, one :app module, minSdk 26



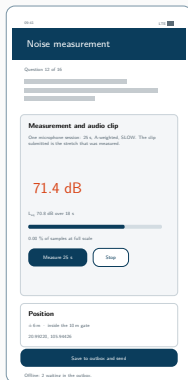
one session: level + clip

# The application



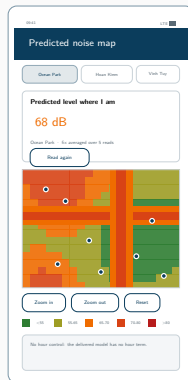
## Collect

both forms, offline



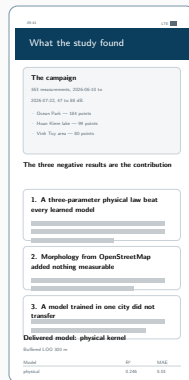
## Measure

25 s, A-weighted



## Map

the level where you stand



## Results

read from metrics.json

Vector mock-ups drawn from the Compose source. No device capture exists yet — which is the last slide of this section.

## What was built — phase by phase

| Phase        | Goal                 | Delivered                                                                                                                                                                           |
|--------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Collect  | Replace ODK Collect  | Both XLSForms as native Compose screens (16 + 6 questions), GPS with the 10 m gate, offline outbox drained by WorkManager, OpenRosa submission                                      |
| 2 — Measure  | Replace Decibel X    | AudioRecord at 48 kHz, A-weighting, SLOW, 25 s window after a discarded warm-up; AGC, noise suppression and echo cancellation <b>off</b> ; the same PCM encoded to AAC for the clip |
| 3 — Show     | Carry the study      | The 40 m grid by hour, the 363 points, the figures, and every metric <i>read</i> from <code>metrics.json</code> — negative results included                                         |
| 4 — Predict  | On-device prediction | The delivered kernel in <code>study/Scenario.kt</code> , under test — and a <b>refusal</b> outside the three mapped areas                                                           |
| 5 — Simulate | Reach the model      | Tier 1 recomputed natively ( $\times 0.2$ to $\times 3$ , $k = 1$ asserted by a test) <i>and</i> the real model driven over <code>gama-server</code>                                |
| 6 — Backend  | Scale                | <b>Not built.</b> Quota, moderation, deduplication, and the path back into <code>01_prepare_field_data.py</code>                                                                    |

5 278 lines of Kotlin, 33 files, 36 JVM unit tests `mobile/PLAN.md`

## Architecture — and the one design rule that matters

|                                          |                               |
|------------------------------------------|-------------------------------|
| app/src/main/java/org/noisehanoi/mobile/ |                               |
| form/                                    | the two XLSForms as data      |
| odk/                                     | instance XML, OpenRosa client |
| outbox/                                  | on-disk queue + WorkManager   |
| measure/                                 | A-weighting, meter, PCM, AAC  |
| location/                                | GPS fixes, the three sites    |
| settings/                                | server config, mic offset     |
| study/                                   | grid, kernel, tier 1 scenario |
| ui/                                      | Compose screens               |

form/, odk/, AWeighting.kt and PcmWriter.kt are **pure Kotlin, no Android dependency** — which is why the constraint logic, the instance format, the weighting curve and the sample byte order are covered by JVM unit tests rather than by an emulator.

### The rule: the app does not carry a copy of the study

syncStudyData in app/build.gradle.kts copies hanoi\_noise\_map.csv, measurements.csv, metrics.json, hybrid\_physical.json and the figures **out of the repository at build time.**

Re-run the pipeline, rebuild the APK, and the app moves with it. No number is transcribed; nothing can drift.

### Same principle, third time

The report reads metrics.json. The dashboard reads metrics.json. Now the app reads metrics.json. **Every published number has exactly one home.**



## Three engineering results worth reporting

**GAMA does not run on a phone — so the app reaches it two ways**

**Tier 1, recomputed natively** over the 5 587 exported cells: a traffic multiplier, offline, within 0.2 dB of the model. **The model itself, over gama-server:** the app draws GAMA's own display and drives it — the scenarios the grid cannot hold, and a server the grid does not need. **Drawing a model's display is not embedding it.**

**The same correction, found a second time**

$$L(k) = 10 \log_{10} (k (E - E_{\text{res}}) + E_{\text{res}}) \quad \text{not} \quad 10 \log_{10}(kE) = L + 10 \log_{10} k$$

The first version took  $E_{\text{res}}$  to be the kernel's constant —  $-99.6$  dB, i.e. nothing — so it scaled the total after all: at  $k = 0.2$ ,  $-7.0$  dB where the model gives  $-3.7$ . **Exactly the pedestrianisation error this project had already corrected**, rebuilt on another platform.  $E_{\text{res}}$  is the zone's own ambience.

**KoboToolbox renames your form at deployment**

The first real submission failed with **404 for a form the server had never heard of**. Kobo discards the XLSForm's `id_string` on deploy and substitutes the asset's own identifier (`aA8FaTuUVSkRjbUW7rCBz7`). The app now reads `/formList` and submits under whatever the server calls the form. **None of this is visible from the phone** — the argument for testing against the real server rather than reasoning about it.

## What the app measures — and what it refuses to claim

**The app's level is a *different quantity* from the campaign's**

This handset's microphone, not a trimmed Decibel X reading. It goes in **its own field**, with the device and the audio source the platform granted. **The two never share a column.**

### Verified end to end, live

- Fix, 25 s measurement, instance written, [submission accepted by kc.kobotoolbox.org](https://kc.kobotoolbox.org)
- A position outside the three areas is **refused, not answered**
- 36 unit tests; weighting pinned to the IEC table

**Not verified — and it is the important one**

**The level has never been checked against a sound source.** An emulator records silence. Until it is compared to a meter, **the absolute figure means nothing** — equally true, per docs/metrology.md, of the campaign's own numbers.

First task on a real handset, alongside Decibel X on the three campaign phones.

# Driving GAMA from the phone — how it is wired

## One socket, and the phone runs nothing

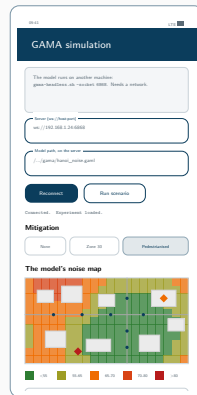
No Android port — but a **WebSocket API**: `gama-headless.sh -socket 6868`. The app is a client (load, play, step, ask, stop): a remote control and a screen.

## Two things only driving it teaches you

A type: `gui` experiment **kills the socket** on load, and step without sync advances nothing while **every indicator reads zero**.

## It redraws the model's display, it does not invent one

Every layer and colour comes from the aspect blocks of `hanoi_noise.gaml`, in **declaration order** — which in **GAMA** is the stacking.



ARCHITECTURE.md §5.2

## Before this app goes to strangers — three blockers

### 1 — Who holds the data

Submissions land in a KoboToolbox account, and whoever owns it holds the positions and timestamps of strangers, and answers for them. It must be **institutional**: a personal account leaves with its owner.

**Not a code change** — the account is a build flag. This is a **decision**, and it is not ours.

`mobile/README.md` · `docs/handover.md §7`

### 2 — Consent is not ethics review

The consent screen exists and is versioned. But a **public campaign is a new collection campaign**, and `handover.md §7` already makes an ethics committee its precondition. The video collection went ahead without review, and the repository says so. **Do not repeat that at a larger scale.**

### 3 — Where does the GAMA server live

Today: whichever laptop runs `gama-headless.sh`, which must be on and reachable. Beyond a demo it needs hosting *and* **TLS** — a release build accepts `wss://` only.

**Next: Clermont-Ferrand**

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### Still no reference instrument

**There will be no sound level meter at Clermont either.** The calibration gap does not close, and nothing said here should suggest it will: the app's level stays relative, and the half-day next to a class 2 meter remains *unbooked*, not *done*.

### So what *is* the work

**Maintenance and deployment**, not new instrumentation. Keep the application alive, then get it collecting in a second city.

## What the second city is actually for

- **One instrument, two cities.** The same **APK** in Hanoi and Clermont. That is the point: the instrument stops being a confound, which is the one thing the Uganda transfer could never fix.
- **Two fabrics as different as we could pick.** Two-wheelers against cars and a tram; dense organic against European planned; flat delta against volcanic relief.
- **A fourth typology, at last.** Three sites bounded everything here. A second city is the only thing that moves that number.

## The comparative deployment — and why it is the decisive test

|            | Hanoi          | Clermont-Ferrand  |
|------------|----------------|-------------------|
| Fleet      | two-wheelers   | cars, tram        |
| Fabric     | dense, organic | European, planned |
| Relief     | flat delta     | volcanic, hilly   |
| Monitoring | none           | EU END mapping    |

**Held constant** — the **same APK**, form, accuracy gate,  $L_{A,25s}$ , OpenStreetMap features and evaluation protocols.

### Why this is stronger than Uganda → Hanoi

There, the **instrument, protocol and feature conventions all differed too**: “transfer fails” and “the datasets are not commensurable” are not separable. One app in both cities and **the instrument stops being a confound**.

### And a ground truth, finally

Clermont-Ferrand falls under EU END strategic mapping — the first **official, instrumented reference map** to compare against.

## The programme, staged

| Stage                 | Work                                                                                                                                                                               | Gate — what must be true to proceed                                                              |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>S1</b><br>Maintain | Keep the application deployable: dependencies, Android target, Kobo form ids, the gama-server protocol. <b>Calibration against a class 1/2 meter stays on this list, unbooked.</b> | The APK still builds and submits, and every level is still reported as relative                  |
| <b>S2</b><br>Repair   | Diagnose the 27 June discontinuity and refit with the phase. Validate the detector. Carry the CIs everywhere.                                                                      | The Hanoi dataset is defensible as published, or its defects are quantified                      |
| <b>S3</b><br>Physics  | CNOSSOS-EU via NoiseModelling + a learned residual, benchmarked against the delivered kernel on the same three splits.                                                             | Real propagation beats two distances and a constant — or it does not, and that is reportable too |
| <b>S4</b><br>Deploy   | Clermont-Ferrand campaign, same APK. Backend for scale (phase 6): quota, moderation, deduplication.                                                                                | A second dataset, commensurable with the first by construction                                   |
| <b>S5</b><br>Compare  | Transfer both ways. Validate against EU END mapping. Publish the two-city methodological study.                                                                                    | —                                                                                                |

Sequenced by dependency, not by calendar. Framing and funding at Clermont-Ferrand are unconfirmed.



## The two things a reviewer will press on

### 1 — No absolute reference. Was half a day really out of reach?

Three handsets calibrated against *each other* and against nothing. docs/metrology.md owns this squarely — but an acoustics reviewer will not ask whether we *knew*; they will ask whether **one side-by-side reading against a class 2 meter** was genuinely unobtainable.

It is the right question, and the honest answer is that a **half-day rental** would settle it. **One reading turns “calibrated in relative terms” into “calibrated”**, and every level inherits it. The cheapest item anywhere in this work.

### 2 — Three sites are three typologies, and that binds harder than the $R^2$

Our own text says it: the number of **independent morphological configurations** is close to **three**, whatever the number of points.  $n = 363$  does not buy  $n = 363$  worth of generalisation.

The hard limit is the design, not the  $R^2$ . **A fourth and fifth contrasting fabric would do more than any modelling change.**

## Conclusion

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# What this work established

## Findings

1. At  $n = 363$  over three sites, **three physical parameters beat every learned model** we built. Capacity does not pay.
2. **Cross-city transfer fails** —  $R^2 < 0$ , with matched features.
3. The video chain **could not be made to work**; that is an instrument result, not an acoustic one.
4. A cross-calibrated smartphone protocol supports **contrasts**, never levels.

## Practices that made those findings trustworthy

- The buffer matches the feature radius — always
- Baselines and an ablation on *the same* splits
- The delivered model is chosen **by code**
- Every number has **exactly one home**
- Retracted work is archived **with its reason**
- Nothing is predicted outside the sampled envelope

The deliverable is a **protocol, an instrument and an honest error bar** — not a noise map of Hanoi.

# Thank you

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`github.com/Colauz/noise-modelling-hanoi`

Everything on these slides rebuilds from the repository:

`make features`    `make models`    `make results`    `make report`

the deck's own figures: `scripts/12_presentation_figures.py`

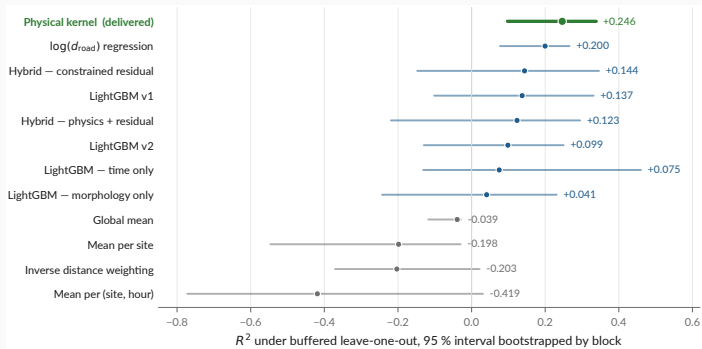
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## Backup — every formula this talk rests on

| What                    | Definition                                                                                                                                                                        |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The measured quantity   | $L_{A,25s} = 10 \log_{10} \left( \frac{1}{T} \int_0^T \frac{p_A^2(t)}{p_0^2} dt \right), \quad T \approx 25 \text{ s}, \quad p_0 = 20 \mu\text{Pa}$                               |
| A-weighting, IEC 61672  | $A(f) = 20 \log_{10} \frac{12194^2 f^4}{(f^2 + 20.6^2) \sqrt{(f^2 + 107.7^2)(f^2 + 737.9^2)(f^2 + 12194^2)}} + 2.00$                                                              |
| Levels never average    | $L_\Sigma = 10 \log_{10} \sum_i 10^{L_i/10}$ decibels add in energy — which is why everything below works on $E$                                                                  |
| The delivered model     | $E = \frac{A_{\text{hw}}}{\max(d_{\text{hw}}, D_0)} + \frac{A_{\text{res}}}{\max(d_{\text{res}}, D_0)} + B, \quad L = 10 \log_{10} E, \quad A_{\text{hw}}, A_{\text{res}}, B > 0$ |
| Why $1/d$ , not $1/d^2$ | A street is a <b>line</b> of sources, not a point: $-3$ dB per doubling of distance, not $-6$                                                                                     |
| The traffic scenario    | $L(k) = 10 \log_{10} (k(E - E_{\text{res}}) + E_{\text{res}})$ <b>not</b> $L + 10 \log_{10} k$                                                                                    |
| Reference protocol      | For each point $i$ , train on $\{j : d(i, j) > r\}$ with $r = 300 \text{ m} =$ <b>the feature aggregation radius</b>                                                              |
| The ceiling on $R^2$    | $\Delta L \sim \mathcal{N}(0, 2\sigma^2) \Rightarrow \sigma = \frac{\sqrt{\pi}}{2} \mathbb{E} \Delta L , \quad R_{\text{max}}^2 = 1 - \sigma^2/s^2$                               |

## Backup — every model, with the interval that qualifies it



`models/metrics.json -- 2000 resamples, bootstrapped by block`

### This is audit finding 3, drawn

physical and dist\_road overlap almost entirely. “The physical kernel beats log-distance” is a **coin flip** at this sample size. Other protocols: `models/model_comparison.md`.

## Backup — the absolute-bias interval

| Anchor                      | Instrument    |   | Quantity                     | Raw gap | Corrected |
|-----------------------------|---------------|---|------------------------------|---------|-----------|
| Gelb & Apparicio 2019, HCMC | dosimeters    | + | $L_{\text{Aeq},1\text{min}}$ | +9.3    | +7.3      |
|                             | GPS           |   |                              |         |           |
| Phan et al. 2010, Hanoi     | RION NL-21/22 |   | $L_{\text{den}}$             | +7.0    | +3.0      |
| Phan et al. 2010, Hanoi     | RION NL-21/22 |   | $L_{\text{night}}$           | −3.0    | −3.0      |
| IOHE, 12 arteries (press)   | undocumented  |   | daytime<br>mean              | +8.6    | +7.6      |

### Conclusion

Plausible absolute bias: **+3.0 to +7.3 dB** (peer-reviewed sources, night excluded —  $n = 10$  is too thin).

**Corrected gap** = raw gap minus the difference the quantity alone explains: it approximates the residual instrumental bias, and **is reported and never applied**.

## Backup — what is published, and what is not

| Dataset                   | Size             | Published | Why / licence        |
|---------------------------|------------------|-----------|----------------------|
| Field measurements        | 363 pts, 3 sites | yes       | CC-BY-4.0            |
| Vehicle counts from video | 147 videos       | yes       | CC-BY-4.0            |
| Survey forms (XLSForm)    | 2 forms          | yes       | CC-BY-4.0            |
| Traffic videos            | 147, 6.0 GB      | no        | faces and plates     |
| Raw Kobo exports          | —                | no        | collector identities |
| OpenStreetMap extract     | 49 146 buildings | no        | regenerable; ODbL    |
| Sunbird Uganda 61K        | 61 k clips       | no        | gated upstream       |

### Stated plainly in the repository

No ethics review was requested for the video collection, and none is claimed. Only non-identifying aggregates are released. The retention decision is still **open**, and is a supervisor decision.

docs/data-sources.md · LICENSE, LICENSE-DATA